Measure Theory

Contents

Problem

Consider a random (infinite) string of 0's and 1's

$$(\ell_1, ..., \ell_n, ...) \quad \ell_n \in \{0, 1\} \forall n \in \mathbb{N}$$

What is the probability that somewhere along the string, I find 100 consecutive zeros?

Comment

The statement of the problem is unclear. Lebesgue's machinery will clarify this

Define

$$X = \{x = (\ell_1, ..., \ell_n, ...) \mid \ell_n \in \{0, 1\} \forall n \in \mathbb{N}\}$$

This is, in a natural way, a compact metric space. Use the "Hamming-style" distance:
For $x = (\ell_1, ..., \ell_n, ...)$ and $x' = (\ell_1{}', ..., \ell_n{}', ...)$

$$d(x, x') = \sum_{n \in \mathbb{N}} \frac{|\ell_n - \ell_n{}'|}{2^n}$$

Since this is a metric space, we can consider the topology:

$$\mathcal{T} = \{G \subseteq X \mid G \text{ is open}\} \subseteq 2^X = \mathcal{P}(X)$$

Can consider

$$\mathfrak{B} = \sigma\text{-Alg}(\mathcal{T}) \quad \text{(Borel sigma algebra)}$$

Then a probability measure is

$$P : \mathcal{B} \longrightarrow [0, 1]$$

Problem 1.1 says:
Let

$$S = \left\{x = (\ell_1, ..., \ell_n, ...) \in X \;\middle|\; \exists m \in \mathbb{N} \text{ with } \ell_m = \ell_{m+1} = \cdots = \ell_{m+99} = 0\right\}$$

check that $S \in \mathcal{B}$, and compute $P(S)$

1 Algebras of Sets

Fix a non-empty set X.

Definition 1.1 **Algebra of Sets**

$\mathfrak{A} \subseteq 2^X$ is an *algebra of sets* if

1. $\varnothing \in \mathfrak{A}$
2. If $A \in \mathfrak{A}$, then $X \setminus A \in \mathfrak{A}$
3. If $A, B \in \mathfrak{A}$, then $A \cap B \in \mathfrak{A}$

Proposition 1.2 — **Properties of algebras of sets**

Let $\mathfrak{A}$ be an algebra of sets on X. Then
1. $X \in \mathfrak{A}$
2. If $A, B \in \mathfrak{A}$, then $A \cup B \in \mathfrak{A}$
3. If $A, B \in \mathfrak{A}$, then $A \setminus B \in \mathfrak{A}$
4. If $\mathfrak{B} \subseteq \mathfrak{A}$ is finite, then $\bigcap \mathfrak{B} \in \mathfrak{A}$
5. If $\mathfrak{B} \subseteq \mathfrak{A}$ is finite, then $\bigcup \mathfrak{B} \in \mathfrak{A}$

Example 1.1

Obviously, 2^X and $\{\varnothing, X\}$ are algebras of sets on X

Lemma 1.3 — **Intersections of algebras**

Let $\mathfrak{F}$ be any collection of algebras of sets on X, then $\bigcap \mathfrak{F}$ is an algebra of sets on X.

Proof: First note if $\mathfrak{F} = \varnothing$, then $\bigcap \mathfrak{F} = 2^X$ which is an algebra of sets on X. We now proceed with checking the axioms of an algebra of sets on X.
1. For every $\mathfrak{A} \in \mathfrak{F}$, we have $\varnothing \in \mathfrak{A}$, thus $\varnothing \in \bigcap \mathfrak{F}$.
2. If $A \in \bigcap \mathfrak{F}$, then $A \in \mathfrak{A}$ for every $\mathfrak{A} \in \mathfrak{F}$. Thus $X \setminus A \in \mathfrak{A}$ for every $\mathfrak{A} \in \mathfrak{F}$, and thus $X \setminus A \in \bigcap \mathfrak{F}$.
3. If $A, B \in \bigcap \mathfrak{F}$, then $A, B \in \mathfrak{A}$ for every $\mathfrak{A} \in \mathfrak{F}$. Thus $A \cap B \in \mathfrak{A}$ for every $\mathfrak{A} \in \mathfrak{F}$, and thus $A \cap B \in \bigcap \mathfrak{F}$.

□

Proposition 1.4 — **Algebra generated by a collection**

Let $\mathfrak{B} \subseteq 2^X$ be any collection of subsets of X, then there exists a unique algebra of sets $\mathfrak{A}$ on X with the following properties:
1. $\mathfrak{B} \subseteq \mathfrak{A}$
2. Whenever $\mathfrak{C}$ is an algebra of sets on X with $\mathfrak{B} \subseteq \mathfrak{C}$, then we have $\mathfrak{A} \subseteq \mathfrak{C}$

Proof: Define $\mathfrak{F}$ to be the collection of all algebras of sets on X that contain $\mathfrak{B}$. Then $\bigcap \mathfrak{F}$ is an algebra of sets by Lemma 1.3 and satisfies (1) and (2). Uniqueness follows from (2). □

Definition 1.5 — **Algebra of sets generated by a collection of subsets**

Let $\mathfrak{B} \subseteq 2^X$ be any collection of subsets of X. We define the *algebra of sets generated by* $\mathfrak{B}$ to be the unique algebra from Proposition 1.4. We notate it as $\operatorname{Alg}(\mathfrak{B})$.

1.1 Additive set-function on an algebra of sets

Definition 1.6 **Additive set-function**

Let $\mathfrak{A} \subseteq 2^X$ be an algebra of sets on X. A function $\mu : \mathfrak{A} \to [0, \infty]$ is called an *additive set-function on* $\mathfrak{A}$ if

1. If $A, B \in \mathfrak{A}$ with $A \cap B = \varnothing$, then $\mu(A \cup B) = \mu(A) + \mu(B)$
2. $\mu(\varnothing) = 0$

Remark 1.7

The arithmetic of $[0, \infty]$ is standard, except for $0 \cdot \infty = \infty \cdot 0 = 0$, which is a useful convention in measure theory.

Remark 1.8

We almost don't need to require (2), it's just when μ is identically ∞ do things go wrong without (2)

Proposition 1.9 **Properties of an additive set function**

Let $\mathfrak{A}$ be an algebra of sets on X, and let μ be an additive set function on $\mathfrak{A}$. Then

1. If $A, C \in \mathfrak{A}$ with $A \subseteq C$, then $\mu(A) \leq \mu(C)$
2. If $A, B \in \mathfrak{A}$, then $\mu(A \cup B) \leq \mu(A) + \mu(B)$ (sub additivity)
3. If $A_1, \ldots, A_n \in \mathfrak{A}$ are all pairwise disjoint, then $\mu(A_1 \cup \cdots \cup A_n) = \mu(A_1) + \cdots + \mu(A_n)$
4. If $A_1, \ldots, A_n \in \mathfrak{A}$, then $\mu(A_1 \cup \cdots \cup A_n) \leq \mu(A_1) + \cdots + \mu(A_n)$

2 Sigma Algebras

Definition 2.1 **Sigma Algebra**

$\mathfrak{M} \subseteq 2^X$ is called a *sigma algebra* on X if

1. $\varnothing \in \mathfrak{M}$
2. If $A \in \mathfrak{M}$, then $X \setminus A \in \mathfrak{M}$
3. If $(A_n)_{n=1}^{\infty} \subseteq \mathfrak{M}$, then $\bigcap_{n=1}^{\infty} A_n \in \mathfrak{M}$

Remark 2.2

countable unions work too, and so do finite operations

Lemma 2.3 **Intersections of sigma algebras**

Let $\mathfrak{F}$ be any collection of sigma algebras on X, then $\bigcap \mathfrak{F}$ is a sigma algebra on X.

Proof: basically identical to the proof for algebras of sets □

Proposition 2.4 **Sigma algebra generated by a collection**

Let $\mathfrak{B} \subseteq 2^X$ be any collection of subsets of X, then there exists a unique sigma algebra $\mathfrak{A}$ on X with the following properties:

1. $\mathfrak{B} \subseteq \mathfrak{A}$
2. Whenever $\mathfrak{C}$ is a sigma algebra on X with $\mathfrak{B} \subseteq \mathfrak{C}$, then we have $\mathfrak{A} \subseteq \mathfrak{C}$

Proof: basically identical to the proof for algebras of sets □

Definition 2.5 **Sigma algebra generated by a collection of subsets**

Let $\mathfrak{B} \subseteq 2^X$ be any collection of subsets of X. We define the *sigma algebra generated by* $\mathfrak{B}$ to be the unique sigma algebra from Proposition 2.4. We notate it as $\sigma\text{-Alg}(\mathfrak{B})$.

Definition 2.6 **Borel Sigma Algebra**

Given a topological space $(X, \mathcal{T})$, the sigma algebra generated by $\mathcal{T}$ is called the *Borel sigma algebra on* X and written $\mathcal{B}_X := \sigma\text{-Alg}(\mathcal{T})$

2.1 Positive measure on a sigma algebra

Remark 2.7 **Increasing limits and series in $[0, \infty]$**

1. Let $(s_n)_{n=1}^{\infty}$ be an *increasing* sequence in $[0, \infty]$ in the sense that

$$s_1 \leq s_2 \leq \cdots$$

   Such a sequence is sure to approach a limit $L \in [0, \infty]$. With a slight abuse of notation we can also write the formula $L = \sup\{s_n \mid n \in \mathbb{N}\}$ meaning that L is the smallest upper bound for all the s_n's. This is usually written in connection to a bounded increasing sequence of numbers in $[0, \infty)$ – our "slight abuse of notation" is that we are also allowing it to cover the case $L = \infty$, which can occur when some of the s_n's are equal to ∞, or when $(s_n)_{n=1}^{\infty}$ is an unbounded sequence in $[0, \infty)$.
2. Let $(t_n)_{n=1}^{\infty}$ be any sequence in $[0, \infty]$. We can consider the finite sums

$$s_n := \sum_{i=1}^{n} t_i$$

   which forms an increasing sequence in $[0, \infty]$. We can consider $L = \lim_{n \to \infty} s_n$ as in (1.), and use the familiar notation $L = \sum_{n=1}^{\infty} t_n$

Definition 2.8 **Positive measure**

Let $\mathfrak{M} \subseteq 2^X$ be sigma algebra of subsets of X. We say $\mu : \mathfrak{M} \to [0, \infty]$ is a *positive measure* on $\mathfrak{M}$ if it satisfies:

1. $\mu(\varnothing) = 0$
2. $$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n)$$

   Whenever $(A_n)_{n=1}^{\infty}$ is a sequence of sets in $\mathfrak{M}$ with the property that if $i, j \in \mathbb{N}$ with $i \neq j$ we have $A_i \cap A_j = \varnothing$

Remark 2.9

A positive measure is, in particular, finitely additive. Thus we can use all the properties in Proposition 1.9

Definition 2.10 **Nomenclature involving measures**

1. If $\mathfrak{M}$ is a sigma algebra on X, then the pair $(X, \mathfrak{M})$ is called a *measurable space*
2. If we also have a positive measure μ on $\mathfrak{M}$, then the triple $(X, \mathfrak{M}, \mu)$ is called a *measure space*
3. If $\mu(X) < \infty$, then we say μ is a *finite measure*
4. If $\mu(X) = 1$, then we say μ is a *probability measure* and that $(X, \mathfrak{M}, \mu)$ is a *probability space*
5. We say that μ is *sigma finite* if there exists a sequence of sets $(X_n)_{n=1}^{\infty}$ in $\mathfrak{M}$ with
   a. $X_1 \subseteq X_2 \subseteq \cdots$ (The X_n's are said to form an increasing chain)
   b. $\bigcup_{n=1}^{\infty} X_n = X$
   c. $\mu(X_n) < \infty$ for all $n \in \mathbb{N}$

Proposition 2.11 **Sigma sub-additivity**

Let $\mathfrak{M}$ be a sigma algebra on X, and let μ be a positive measure on $\mathfrak{M}$. If $(A_n)_{n=1}^{\infty}$ is a sequence of sets in $\mathfrak{M}$, then

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu(A_n)$$

Proposition 2.12 **Continuity along chains**

Let $\mathfrak{M}$ be a sigma algebra on X, and let μ be a positive measure on $\mathfrak{M}$.

1. If $(A_n)_{n=1}^{\infty}$ is an increasing chain in $\mathfrak{M}$, then
$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} \mu(A_n)$$
2. If $(A_n)_{n=1}^{\infty}$ is a decreasing chain in $\mathfrak{M}$ with $\mu(A_1) < \infty$, then
$$\mu\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} \mu(A_n)$$

Example 2.1

Example of "atomic" μ based on weights of individual points $x \in X$. Suppose you have $w : X \to [0, \infty]$. Let $\mathfrak{M} = 2^X$, for every $A \subseteq X$, put

$$\mu(A) = \sup\left\{\sum_{x\in F} w(x) \;\middle|\; F \subseteq A \text{ finite}\right\}$$

Then $(X, \mathfrak{M}, \mu)$ is a measure space. For the special case that $w(x) = 1, \forall x \in X$ is called the *counting measure* on X. For the special case that $w(x_0) = 1$ for some $x_0 \in X$ and $w(x) = 0, \forall x \neq x_0$ is called the *Dirac measure concentrated at* x_0

2.2 A glimpse of the Lebesgue measure on $\mathbb{R}$

Note

1. $\mathcal{B}_{\mathbb{R}}$ can be generated from the half open intervals
2. If $\mu, \nu : \mathcal{B}_{\mathbb{R}} \to [0, \infty]$ are positive measures such that
$$\mu((a, b]) = \nu((a, b]) < \infty, \forall a < b \in \mathbb{R}$$
   It follows that $\mu = \nu$

Then the *Lebesgue measure* $\lambda : \mathcal{B}_{\mathbb{R}} \to [0, \infty]$ is determined via with requirement that $\lambda((a, b]) = b - a, \forall a < b \in \mathbb{R}$.
Another approach: λ is uniquely determined by the requirement that $\lambda(B + t) = \lambda(B)$ for every $B \in \mathcal{B}_{\mathbb{R}}$ and $t \in \mathbb{R}$ with the normalization that $\lambda((0, 1]) = 1$.

Note

For the *existence* of λ, will invoke an extension theorem of Caratheodory (to be done later)

Note

Interesting fact: there is no positive measure $\mu : 2^{\mathbb{R}} \to [0, \infty]$ which is translation invariant and $\mu((0, 1]) = 1$. This shows, in particular that $\mathcal{B}_{\mathbb{R}} \neq 2^{\mathbb{R}}$

3 Measurable functions

Definition 3.1 **Measurable function**

Let $(X, \mathfrak{M})$, $(Y, \mathfrak{N})$ be measurable spaces and $f : X \to Y$. Then f is called $\mathfrak{M}/\mathfrak{N}$-*measurable* if $f^{-1}(N) \in \mathfrak{M}$ for every $N \in \mathfrak{N}$.

Remark 3.2

Pre-images behave nicely with respect to the usual set operations.

Proposition 3.3 **Composition of measurable maps**

Compositions of measurable maps are measurable.

Proof: Obvious □

Proposition 3.4 **Measurability criterion**

Let $(X, \mathfrak{M})$, $(Y, \mathfrak{N})$ be measurable spaces. $f : X \to Y$, and $\mathfrak{C} \subseteq \mathfrak{N}$ such that $\sigma\text{-Alg}(\mathfrak{C}) = \mathfrak{N}$ and such $f^{-1}(C) \in \mathfrak{M}$ for every $C \in \mathfrak{C}$. Then f is measurable.

Proof: Let $\mathfrak{Q} = \{Q \subseteq Y \mid f^{-1}(Q) \in \mathfrak{M}\} \subseteq 2^Y$

Claim: $\mathfrak{Q}$ is a sigma algebra

Note by assumption $\mathfrak{C} \subseteq \mathfrak{Q}$ thus $\mathfrak{N} = \sigma\text{-Alg}(\mathfrak{C}) \subseteq \mathfrak{Q}$. Thus f is measurable. □

Corollary 3.5 **Continuous maps are measurable**

Let X, Y be topological spaces, $f : X \to Y$ continuous. Then f is $\mathcal{B}_X/\mathcal{B}_Y$-measurable.

3.1 The space of functions $\mathrm{Bor}(X, \mathbb{R})$

Definition 3.6 **Borel measurable real-valued functions**

Given a measurable space $(X, \mathfrak{M})$. Define

$$\mathrm{Bor}(X, \mathbb{R}) := \{f : X \to \mathbb{R} \mid f \text{ is } \mathfrak{M}/\mathcal{B}_{\mathbb{R}} - \text{measurable}\}$$

Proposition 3.7 **Algebra and lattice operations**

Let $(X, \mathfrak{M})$ be a measurable space with $f, g \in \text{Bor}(X, \mathbb{R})$. Then

1. $f + g \in \text{Bor}(X, \mathbb{R})$
2. For any $c \in \mathbb{R}$, $cf \in \text{Bor}(X, \mathbb{R})$
3. $f \cdot g \in \text{Bor}(X, \mathbb{R})$
4. $f \vee g, f \wedge g \in \text{Bor}(X, \mathbb{R})$

Proof of 3: Let $f, g \in \text{Bor}(X, \mathbb{R})$. Let $h : X \to \mathbb{R}^2$ given by $h(x) = (f(x), g(x))$. Let $p : \mathbb{R}^2 \to \mathbb{R}$ given by $p(s, t) = s \cdot t$.

Claim: h is measurable

Claim: p is measurable

Claim: $f \cdot g = p \circ h$

Thus $f \cdot g \in \text{Bor}(X, \mathbb{R})$ □

Exercise 3.1

Let $\mathfrak{S}$ be the collection of open squares in $\mathbb{R}^2$. Then $\sigma\text{-Alg}(\mathfrak{S}) = \mathcal{B}_{\mathbb{R}^2}$.

Remark 3.8

Let $(X, \mathfrak{M})$ be a measurable space. Then $\text{Bor}(X, \mathbb{R})$ is an *algebra of functions* and a *lattice of functions*.

Remark 3.9

Operations with only one function. If $f \in \text{Bor}(X, \mathbb{R})$, then $|f| \in \text{Bor}(X, \mathbb{R})$. Use algebraic stability properties to prove this.

Example 3.1 **Problem 3.1**

Claim: Let $\mathfrak{C} = \{(-\infty, t], \mid t \in \mathbb{R}\} \subseteq \mathcal{B}_{\mathbb{R}}$. Then $\sigma\text{-Alg}(\mathfrak{C}) = \mathcal{B}_{\mathbb{R}}$.

Proof: Denote $\sigma\text{-Alg}(\mathfrak{C}) = \mathfrak{P}$. Clearly, $\mathfrak{P} \subseteq \mathcal{B}_{\mathbb{R}}$. If we can show that $G \in \mathfrak{P}$ for all $G \subseteq \mathbb{R}$ open, then we will be done because we will have that $\mathfrak{P}$ is a sigma-algebra and that $\mathfrak{P} \supseteq \mathcal{T}_{\mathbb{R}}$.
Now, every $\varnothing \neq G \subseteq \mathbb{R}$ open can be written as a countable union of open intervals. So then, it suffices to check that $\mathfrak{P}$ contains all of the open intervals of $\mathbb{R}$. This is left as an exercise. □

Let $(X, \mathfrak{M})$ be a measurable space, and suppose $f : X \to \mathbb{R}$ satisfies $f^{-1}((-\infty, t]) \in \mathfrak{M}$ for all $t \in \mathbb{R}$. That is, $f^{-1}(C) \in \mathfrak{M}$ for all $C \in \mathfrak{C}$. Then by Proposition 3.4, $f^{-1}(B) \in \mathfrak{M}$ for all $B \in \mathcal{B}_{\mathbb{R}}$, hence $f \in \text{Bor}(X, \mathbb{R})$.

4 Convergence properties of $\mathrm{Bor}(X, \mathbb{R})$

Proposition 4.1 **Closure under suprema and infima**

Let $(X, \mathfrak{M})$ be a measurable space.

1. Let $f : X \to \mathbb{R}$ be a function. Let $(f_n)_{n=1}^{\infty}$ be a sequence of functions in $\mathrm{Bor}(X, \mathbb{R})$ such that
$$\sup\{f_n(x) \mid n \in \mathbb{N}\} = f(x)$$
for all $x \in X$. Then it follows that $f \in \mathrm{Bor}(X, \mathbb{R})$.
2. Let $g : X \to \mathbb{R}$ be a function. Let $(g_n)_{n=1}^{\infty}$ be a sequence of functions in $\mathrm{Bor}(X, \mathbb{R})$ such that
$$\inf\{g_n(x) \mid n \in \mathbb{N}\} = g(x)$$
for all $x \in X$. Then it follows that $g \in \mathrm{Bor}(X, \mathbb{R})$.

Proof:

1. It suffices to show that $f^{-1}((-\infty, t]) \in \mathfrak{M}$ for all $t \in \mathbb{R}$. This is left as an exercise.
2. Exercise.

□

Remark 4.2

Let $(t_n)_{n=1}^{\infty}$ be a bounded sequence in $\mathbb{R}$. Then there exists $\ell_+ \in \mathbb{R}$ such that

1. There exists a subsequence of $(t_n)_{n=1}^{\infty}$ which converges to ℓ_+.
2. There is no subsequence of $(t_n)_{n=1}^{\infty}$ which converges to an $\ell > \ell_+$.

Define
$$\limsup_{n\to\infty} t_n := \ell_+$$
We can obtain ℓ_+ as
$$\ell_+ = \lim_{k\to\infty}\left(\sup_{n\geq k} t_n\right) = \inf_{k\geq 1}\left(\sup_{n\geq k} t_n\right)$$
We can define $\liminf_{n\to\infty} t_n$ similarly.

Fix a measurable space $(X, \mathfrak{M})$

Proposition 4.3

1. Let $f : X \to \mathbb{R}$. Suppose we could find $f_1, f_2, \ldots \in \mathrm{Bor}(X, \mathbb{R})$ such that for every $x \in X$, the sequence $(f_n(x))_{n=1}^{\infty}$ is bounded with
$$\limsup_{n \to \infty} f_n(x) = f(x)$$
Then it follows that $f \in \mathrm{Bor}(X, \mathbb{R})$
2. Let $g : X \to \mathbb{R}$. Suppose we could find $g_1, g_2, \ldots \in \mathrm{Bor}(X, \mathbb{R})$ such that for every $x \in X$, the sequence $(g_n(x))_{n=1}^{\infty}$ is bounded with
$$\liminf_{n \to \infty} g_n(x) = g(x)$$
Then it follows that $g \in \mathrm{Bor}(X, \mathbb{R})$

Proof of 1: For every $k \in \mathbb{N}$, define the helper function $h_k : X \to \mathbb{R}$ by
$$h_k(x) = \sup \underbrace{\{f_n(x) \mid n \geq k\}}_{\text{bounded by hypothesis}}, \ \forall x \in X$$
Observe $f_k, f_{k+1}, \ldots \in \mathrm{Bor}(X, \mathbb{R})$. We can invoke Proposition 4.1, to conclude that $h_k \in \mathrm{Bor}(X, \mathbb{R})$. Observe for every $x \in X$, we have
$$h_1(x) \geq h_2(x) \geq \cdots$$
with
$$f(x) = \lim_{k \to \infty} h_k(x) = \inf\{h_k(x) \mid k \in \mathbb{N}\}$$
This is because $f(x) = \limsup_{n \to \infty} f_n(x)$ by hypothesis. But then
$$h_k \in \mathrm{Bor}(X, \mathbb{R}), \ \forall k \in \mathbb{N}$$
and
$$f(x) = \inf\{h_k(x) \mid k \in \mathbb{N}\} \ \forall x \in X$$
thus $f \in \mathrm{Bor}(X, \mathbb{R})$ by Proposition 4.1. □

Proof of 2: We can reduce to (1), by introducing $f : X \to \mathbb{R}$, $f(x) = -g(x) \ \forall x \in X$ and introducing $f_n = -g_n \in \mathrm{Bor}(X, \mathbb{R}), \forall n \in \mathbb{N}$, then apply (1) to $f(x) = \limsup_{n \to \infty} f_n(x)$, to get $f \in \mathrm{Bor}(X, \mathbb{R})$, hence $g \in \mathrm{Bor}(X, \mathbb{R})$. □

Corollary 4.4

Let $f : X \to \mathbb{R}$. Suppose we could find $(f_n)_{n=1}^{\infty}$ in $\mathrm{Bor}(X, \mathbb{R})$ such that $f(x) = \lim_{n \to \infty} f_n(x), \ \forall x \in X$. Then it follows that $f \in \mathrm{Bor}(X, \mathbb{R})$.

Proof: Use Proposition 4.3. □

Exercise 4.1

Suppose that $(X, \mathfrak{M}) = (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$. Let $f : \mathbb{R} \to \mathbb{R}$ be differentiable and consider the function $f' : \mathbb{R} \to \mathbb{R}$ (f' may not be continuous).

Solution:
Idea is to produce $(f_n)_{n=1}^{\infty}$ in $\mathrm{Bor}(\mathbb{R}, \mathbb{R})$ such that $\lim_{n\to\infty} f_n(x) = f'(x)$. If we find the f_n's then $f' \in \mathrm{Bor}(\mathbb{R}, \mathbb{R})$ due to Corollary 4.4. How do we find the f_n's? For $x \in \mathbb{R}$

$$f'(x) = \lim_{n\to\infty} \frac{f\left(x + \frac{1}{n}\right) - f(x)}{\frac{1}{n}} = \lim_{n\to\infty} n(u_n(x) - f(x))$$

With $u_n(x) = f\left(x + \frac{1}{n}\right)$. So then, for every $n \in \mathbb{N}$, let $u_n : \mathbb{R} \to \mathbb{R}$ be defined by $u_n(x) = f\left(x + \frac{1}{n}\right),\ \forall x \in \mathbb{R}$ and let $f_n = n(u_n - f)$. Every f_n is continuous, hence in $\mathrm{Bor}(\mathbb{R}, \mathbb{R})$ by Corollary 3.5 and $\lim_{n\to\infty} f_n(x) = f'(x)$ as we wanted.

4.1 Approximation by simple functions

Definition 4.5 **Simple functions**

A function $f \in \mathrm{Bor}(X, \mathbb{R})$ is said to be *simple* when it takes finitely many values, that is $f(X)$ is a finite subset of $\mathbb{R}$. We denote

$$\mathrm{Bor}_s(X, \mathbb{R}) := \{f \in \mathrm{Bor}(X, \mathbb{R}) \mid f \text{ is simple}\}$$

We also introduce the spaces

$$\mathrm{Bor}^+(X, \mathbb{R}) = \{f \in \mathrm{Bor}(X, \mathbb{R}) \mid f(x) \geq 0\ \forall x \in X\}$$

and

$$\mathrm{Bor}_s^+(X, \mathbb{R}) := \mathrm{Bor}_s(X, \mathbb{R}) \cap \mathrm{Bor}^+(X, \mathbb{R})$$

Note

For $A \in \mathfrak{M}$, we have $\chi_A \in \mathrm{Bor}_s(X, \mathbb{R})$, where $\chi_A : X \to \mathbb{R}$ is defined by

$$\chi_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \in X \setminus A \end{cases}$$

called the *indicator function of A*. Also note that χ_A is $\mathfrak{M}/\mathcal{B}_{\mathbb{R}}$-measurable since for every $S \in \mathcal{B}_{\mathbb{R}}$ we have that

$$\chi_A^{-1}(S) \in \{\varnothing, A, X \setminus A, X\} \subseteq \mathfrak{M}$$

Note

It can be shown that

$$\mathrm{Bor}_s(X,\mathbb{R}) = \underbrace{\mathrm{span}\{\chi_A \mid A \in \mathfrak{M}\}}_{\substack{\text{set of all linear combinations} \\ \text{of indicator functions} \\ \chi_A \text{ with } A\in\mathfrak{M}}}$$

Remark 4.6 **Binning values of a non-negative Borel function**

Let $f \in \mathrm{Bor}^+(X,\mathbb{R})$. For every $n \in \mathbb{N}$, let $f_n : X \to \mathbb{R}$ be the *n-th binning of the values of f*, defined as follows. For each $x \in X$:

1. If $f(x) \geq n$, define $f_n(x) := n$.
2. If $f(x) < n$, pick the unique $k \in \{1, \ldots, n \cdot 2^n\}$ such that

$$\frac{k-1}{2^n} \leq f(x) < \frac{k}{2^n}$$

   and define $f_n(x) := \frac{k-1}{2^n}$.

Proposition 4.7 **Binning approximation**

Let $f \in \mathrm{Bor}^+(X,\mathbb{R})$, and for every $n \in \mathbb{N}$, let f_n be the n-th binning of the values of f. Then

1. $f_n \in \mathrm{Bor}_s^+(X,\mathbb{R})$ for every $n \in \mathbb{N}$
2. $f_n \leq f_{n+1}$ for every $n \in \mathbb{N}$
3. For every $x \in X$,

$$\lim_{n\to\infty} f_n(x) = f(x)$$

5 Integration of functions in $\mathrm{Bor}^+(X,\mathbb{R})$

5.1 The map $L_s^+ : \mathrm{Bor}_s^+(X,\mathbb{R}) \longrightarrow [0,\infty]$

Fix a measure space $(X,\mathfrak{M},\mu)$

Note

For didactical reasons, it is convenient to first define $\int_X f(x)\,\mathrm{d}\mu(x)$ for $f \in \mathrm{Bor}^+(X,\mathbb{R})$. In this case, the integral is always defined but may be ∞. For general $f \in \mathrm{Bor}(X,\mathbb{R})$, the integral may not be defined; to get a finite integral, we need $f \in \mathcal{L}^1(\mu)$.

Moreover, it will be convenient to introduce the integral on $\mathrm{Bor}^+(X,\mathbb{R})$ as a map

$$L^+ : \mathrm{Bor}^+(X,\mathbb{R}) \longrightarrow [0,\infty].$$

We will get L^+ in two parts:

1. Define $L_s^+ : \mathrm{Bor}_s^+(X,\mathbb{R}) \to [0,\infty]$
2. Extend L_s^+ to a map $L^+ : \mathrm{Bor}^+(X,\mathbb{R}) \to [0,\infty]$

Remark 5.1 **Easily verified properties of $\mathrm{Bor}_s^+(X,\mathbb{R})$**

1. If $f, g \in \mathrm{Bor}_s^+(X,\mathbb{R})$, then $f+g \in \mathrm{Bor}_s^+(X,\mathbb{R})$
2. If $f \in \mathrm{Bor}_s^+(X,\mathbb{R})$ and $\alpha \in [0,\infty)$, then $\alpha f \in \mathrm{Bor}_s^+(X,\mathbb{R})$
3. If $A \in \mathfrak{M}$, then $\chi_A \in \mathrm{Bor}_s^+(X,\mathbb{R})$

Note

As a consequence of (3), taking $A = \varnothing$ and $A = X$, we get $\underline{1}, \underline{0} : X \to \mathbb{R} \in \mathrm{Bor}_s^+(X,\mathbb{R})$.

Theorem 5.2 **Integral of non-negative simple functions**

There exists a map

$$L_s^+ : \mathrm{Bor}_s^+(X,\mathbb{R}) \longrightarrow [0,\infty],$$

uniquely determined, such that the following conditions are satisfied:

(i) Additivity:

$$L_s^+(f+g) = L_s^+(f) + L_s^+(g)$$

for all $f, g \in \mathrm{Bor}_s^+(X,\mathbb{R})$

(ii) Positive homogeneity:

$$L_s^+(\alpha f) = \alpha L_s^+(f)$$

for all $f \in \mathrm{Bor}_s^+(X,\mathbb{R})$ and $\alpha \in [0,\infty)$

(iii) Relation to the measure:

$$L_s^+(\chi_A) = \mu(A)$$

for all $A \in \mathfrak{M}$

In order to prove the existence of the map L_s^+, we will take advantage of the fact that one can describe the functions from $\mathrm{Bor}_s^+(X,\mathbb{R})$ in terms of measurable divisions of X.

Definition 5.3 **Measurable division and intersections of measurable divisions**

1. A *measurable division* of X is a set $\{A_1, \ldots, A_n\} \subseteq \mathfrak{M}$ of pairwise disjoint, non-empty sets such that $\bigcup_{i=1}^n A_i = X$
2. Given two measurable divisions $\Delta = \{A_1, \ldots, A_n\}$ and $\Gamma = \{B_1, \ldots, B_m\}$ of X, one can "intersect" them, to form a new measurable division of X, denoted $\Delta \wedge \Gamma$, defined by

$$\Delta \wedge \Gamma := \left\{ A_i \cap B_j \;\middle|\; 1 \leq i \leq n, 1 \leq j \leq m, \text{ and } A_i \cap B_j \neq \varnothing \right\}$$

Lemma 5.4 **Simple functions and measurable divisions**

Let $f \in \mathrm{Bor}_s^+(X, \mathbb{R})$.

1. There exists a measurable division $\Delta = \{A_1, ..., A_n\}$ of X and numbers $\alpha_1, ..., \alpha_n \in [0, \infty)$ such that
$$f = \alpha_1 \chi_{A_1} + \cdots + \alpha_n \chi_{A_n}$$
2. Suppose that $\Delta = \{A_1, ..., A_n\}$ and $\alpha_1, ..., \alpha_n \in [0, \infty)$ are as in (1.), and suppose that $\Gamma = \{B_1, ..., B_m\}$ is a second measurable division of X. Then
$$f = \sum_{\substack{1 \le i \le n, \\ 1 \le j \le m, \\ A_i \cap B_j \ne \varnothing}} \alpha_i \chi_{A_i \cap B_j}$$

Proof of 2: For every $1 \le i \le n$, write

$$\chi_{A_i} = \chi_{A_i} \cdot 1 = \chi_{A_i}\left(\chi_{B_1} + \cdots + \chi_{B_m}\right) = \sum_{j=1}^{m} \chi_{A_i \cap B_j} = \sum_{\substack{1 \le j \le m, \\ A_i \cap B_j \ne \varnothing}} \chi_{A_i \cap B_j}$$

Then

$$f = \sum_{i=1}^{n} \alpha_i \chi_{A_i} = \sum_{i=1}^{n} \alpha_i \left(\sum_{\substack{1 \le j \le m, \\ A_i \cap B_j \ne \varnothing}} \chi_{A_i \cap B_j} \right) = \sum_{\substack{1 \le i \le n, \\ 1 \le j \le m, \\ A_i \cap B_j \ne \varnothing}} \alpha_i \chi_{A_i \cap B_j}$$

as desired. □

Lemma 5.5 **Well-definedness of the simple integral formula**

Let $f \in \mathrm{Bor}_s^+(X, \mathbb{R})$. Suppose we have two ways of writing f as a linear combination:

$$f = \alpha_1 \chi_{A_1} + \cdots + \alpha_n \chi_{A_n}$$

and

$$f = \beta_1 \chi_{B_1} + \cdots + \beta_m \chi_{B_m},$$

where $\Delta = \{A_1, ..., A_n\}$ and $\Gamma = \{B_1, ..., B_m\}$ are measurable divisions of X, and $\alpha_1, ..., \alpha_n, \beta_1, ..., \beta_m \in [0, \infty)$. Then

$$\sum_{i=1}^{n} \alpha_i \mu(A_i) = \sum_{j=1}^{m} \beta_j \mu(B_j) \in [0, \infty].$$

Proof: For every $1 \le i \le n$, let us write

$$A_i = A_i \cap X = A_i \cap (B_1 \cup \cdots \cup B_m) = \bigcup_{j=1}^{m} (A_i \cap B_j) = \bigcup_{\substack{1 \leq j \leq m, \\ A_i \cap B_j \neq \varnothing}} (A_i \cap B_j)$$

Since the union is disjoint,

$$\mu(A_i) = \sum_{\substack{1 \leq j \leq m, \\ A_i \cap B_j \neq \varnothing}} \mu(A_i \cap B_j)$$

so

$$\sum_{i=1}^{n} \alpha_i \mu(A_i) = \sum_{i=1}^{n} \alpha_i \sum_{\substack{1 \leq j \leq m, \\ A_i \cap B_j \neq \varnothing}} \mu(A_i \cap B_j) = \sum_{\substack{1 \leq i \leq n, \\ 1 \leq j \leq m, \\ A_i \cap B_j \neq \varnothing}} \alpha_i \mu(A_i \cap B_j)$$

Similarly,

$$\sum_{j=1}^{m} \beta_j \mu(B_j) = \sum_{j=1}^{m} \beta_j \sum_{\substack{1 \leq i \leq n, \\ B_j \cap A_i \neq \varnothing}} \mu(B_j \cap A_i) = \sum_{\substack{1 \leq i \leq n, \\ 1 \leq j \leq m, \\ A_i \cap B_j \neq \varnothing}} \beta_j \mu(A_i \cap B_j)$$

Claim: If $A_i \cap B_j \neq \varnothing$, then $\alpha_i = \beta_j$

Observe that this claim will finish the proof. To verify the claim, suppose $A_i \cap B_j \neq \varnothing$. Pick $x \in A_i \cap B_j$, and evaluate $f(x)$ in two ways:

$$f(x) = \alpha_1 \chi_{A_1}(x) + \cdots + \alpha_n \chi_{A_n}(x) = \alpha_i$$

similarly

$$f(x) = \beta_1 \chi_{B_1}(x) + \cdots + \beta_m \chi_{B_m}(x) = \beta_j,$$

so $\alpha_i = f(x) = \beta_j$ □

Proof of existence of L_s^+ from Theorem 5.2: For every $f \in \mathrm{Bor}_s^+(X, \mathbb{R})$, we define the quantity $L_s^+(f) \in [0, \infty]$ by the following procedure:

Consider a writing

$$f = \alpha_1 \chi_{A_1} + \cdots + \alpha_n \chi_{A_n}$$

$$\text{with } \Delta = \{A_1, ..., A_n\} \text{ a measurable division of } X \text{ and } \alpha_1, ..., \alpha_n \in [0, \infty), \tag{$*$}$$

and put

$$L_s^+(f) := \sum_{i=1}^{n} \alpha_i \mu(A_i) \in [0, \infty].$$

Lemma 5.5 assures us that the procedure $(*)$ is coherent, in the sense that the proposed quantity $L_s^+(f)$ only depends on f and not on the choice of how we write f as a linear combination of indicators.

With $(*)$, we have thus defined a map

$$L_s^+ : \mathrm{Bor}_s^+(X, \mathbb{R}) \longrightarrow [0, \infty].$$

We must now check that this map has the properties (i), (ii), (iii) indicated in Theorem 5.2. □

5.2 The map $L^+ : \mathrm{Bor}^+(X, \mathbb{R}) \longrightarrow [0, \infty]$

Definition 5.6 **Integral of non-negative Borel functions**

For every function $u \in \mathrm{Bor}^+(X, \mathbb{R})$, we define

$$L^+(u) := \sup\{L_s^+(f) \mid f \in \mathrm{Bor}_s^+(X, \mathbb{R}), f \leq u\} \in [0, \infty].$$

Remark 5.7 **Comments on the formula for L^+**

- We use letters such as $u, v, \ldots$ for general functions in $\mathrm{Bor}^+(X, \mathbb{R})$, reserving $f, g, \ldots$ for functions in $\mathrm{Bor}_s^+(X, \mathbb{R})$.
- In the formula above, $f \leq u$ means $f(x) \leq u(x)$ for every $x \in X$.
- The set over which we take the supremum is non-empty because it contains $\underline{0}$.
- The supremum is taken in $[0, \infty]$, so it is allowed to be ∞.

Proposition 5.8 **L^+ extends L_s^+**

L^+ is an extension of L_s^+. That is, if $u \in \mathrm{Bor}_s^+(X, \mathbb{R})$, then the newly defined quantity $L^+(u)$ is the same as $L_s^+(u)$.

Proof: Must show

$$\sup\{L_s^+(f) \mid f \in \mathrm{Bor}_s^+(X, \mathbb{R}), f \leq u\} = L_s^+(u)$$

"$\geq$" holds because L_s^+ is an element of $\{L_s^+(f) \mid f \in \mathrm{Bor}_s^+(X, \mathbb{R}), f \leq u\}$. For "$\leq$", we must show $L_s^+(u)$ is an upper bound for $\{L_s^+(f) \mid f \in \mathrm{Bor}_s^+(X, \mathbb{R}), f \leq u\}$. Indeed, if $f \in \mathrm{Bor}_s^+(X, \mathbb{R})$ with $f \leq u$, then, using Theorem 5.2,

$$L_s^+(u) = L_s^+(f + (u - f)) = L_s^+(f) + L_s^+(u - f) \geq L_s^+(f)$$

□

Proposition 5.9 **Monotonicity and homogeneity of L^+**

1. L^+ is an increasing map: if $u, v \in \mathrm{Bor}^+(X, \mathbb{R})$ with $u \leq v$, then $L^+(u) \leq L^+(v)$
2. L^+ is homogeneous: if $u \in \mathrm{Bor}^+(X, \mathbb{R})$, $\alpha \in [0, \infty)$, then $L^+(\alpha u) = \alpha L^+(u)$

Proof of 1: Since $u \leq v$, we have

$$\{f \in \mathrm{Bor}_s^+(X, \mathbb{R}) \mid f \leq u\} \subseteq \{f \in \mathrm{Bor}_s^+(X, \mathbb{R}) \mid f \leq v\}.$$

Thus

$$\begin{aligned} L^+(u) &= \sup\{L_s^+(f) \mid f \in \mathrm{Bor}_s^+(X,\mathbb{R}), f \le u\} \\ &\le \sup\{L_s^+(f) \mid f \in \mathrm{Bor}_s^+(X,\mathbb{R}), f \le v\} \\ &= L^+(v). \end{aligned}$$

□

Proof of 2: If $\alpha = 0$, then $L^+(\alpha u) = 0 = \alpha L^+(u)$, where we use the convention $0 \cdot \infty = 0$.

Suppose $\alpha > 0$. Then

$$\{g \in \mathrm{Bor}_s^+(X,\mathbb{R}) \mid g \le \alpha u\} = \{\alpha f \mid f \in \mathrm{Bor}_s^+(X,\mathbb{R}), f \le u\}.$$

Thus,

$$\begin{aligned} L^+(\alpha u) &= \sup\{L_s^+(\alpha f) \mid f \in \mathrm{Bor}_s^+(X,\mathbb{R}), f \le u\} \\ &= \sup\{\alpha L_s^+(f) \mid f \in \mathrm{Bor}_s^+(X,\mathbb{R}), f \le u\} \\ &= \alpha \sup\{L_s^+(f) \mid f \in \mathrm{Bor}_s^+(X,\mathbb{R}), f \le u\} \\ &= \alpha L^+(u), \end{aligned}$$

where the second equality uses Theorem 5.2. □

Remark 5.10 **Additivity and MCT**

L^+ is additive: $L^+(u+v) = L^+(u) + L^+(v)$ for every $u, v \in \mathrm{Bor}^+(X,\mathbb{R})$. We can show $L^+(u+v) \ge L^+(u) + L^+(v)$ by an easy argument using the definition, but the other inequality is not so easy. Instead we move to the Monotone Convergence Theorem (MCT), which will prove it. Note we have to make sure not to use additivity in the proof of MCT.

6 The Monotone Convergence Theorem

Fix a measure space $(X, \mathfrak{M}, \mu)$ from chapter 5. We will work with the map $L^+ : \mathrm{Bor}^+(X,\mathbb{R}) \to [0,\infty]$.

Notation 6.1 **Increasing convergence**

For u and $u_1, u_2, \ldots \in \mathrm{Bor}^+(X,\mathbb{R})$, we will use the shorthand $u_n \nearrow u$ to mean that $u_1 \le u_2 \le \cdots$ and $u_n \to u$ pointwise.

Theorem 6.2 **Monotone Convergence Theorem**

Let $u, u_1, u_2, \ldots \in \mathrm{Bor}^+(X,\mathbb{R})$ with $u_n \nearrow u$. Then

$$\lim_{n\to\infty} L^+(u_n) = L^+(u)$$

(an increasing limit in $[0,\infty]$)

Proposition 6.3 **The missing inequality for additivity of L^+**

Let $u, v \in \operatorname{Bor}^+(X, \mathbb{R})$. Then

$$L^+(u+v) \leq L^+(u) + L^+(v).$$

Proof: By Proposition 4.7, choose sequences $(f_n)_{n=1}^\infty$ and $(g_n)_{n=1}^\infty$ in $\operatorname{Bor}_s^+(X, \mathbb{R})$ such that $f_n \nearrow u$ and $g_n \nearrow v$. It follows that $(f_n + g_n)_{n=1}^\infty$ is a sequence in $\operatorname{Bor}_s^+(X, \mathbb{R})$ such that $f_n + g_n \nearrow u + v$.

Thus,

$$\begin{aligned} L^+(u+v) &= \lim_{n\to\infty} L^+(f_n + g_n) \\ &= \lim_{n\to\infty} L_s^+(f_n + g_n) \\ &= \lim_{n\to\infty} (L_s^+(f_n) + L_s^+(g_n)) \\ &= \lim_{n\to\infty} (L^+(f_n) + L^+(g_n)). \end{aligned}$$

The first equality uses Theorem 6.2, the second and fourth use Proposition 5.8, and the third uses Theorem 5.2.

Now for every $n \in \mathbb{N}$, since $f_n \leq u$ and $g_n \leq v$, Proposition 5.9 gives

$$L^+(f_n) + L^+(g_n) \leq L^+(u) + L^+(v).$$

Hence

$$L^+(u+v) \leq L^+(u) + L^+(v),$$

as desired. □

6.1 Proof of the Monotone Convergence Theorem

Remark 6.4

If $u_n \nearrow u$ in $\mathrm{Bor}^+(X, \mathbb{R})$, then

$$\Lambda := \lim_{n\to\infty} L^+(u_n) = \sup\{L^+(u_n) \mid n \in \mathbb{N}\} \qquad \text{(point-to-prove)}$$

will satisfy the inequality $\Lambda \geq L^+(u)$.

To see that this is enough to prove Theorem 6.2, suppose $u_n \nearrow u$ in $\mathrm{Bor}^+(X, \mathbb{R})$. Since $u_n \leq u$ for every $n \in \mathbb{N}$, Proposition 5.9 gives

$$L^+(u_n) \leq L^+(u)$$

for every $n \in \mathbb{N}$. Hence $L^+(u)$ is an upper bound for $\{L^+(u_n) \mid n \in \mathbb{N}\}$, so

$$\Lambda = \sup\{L^+(u_n) \mid n \in \mathbb{N}\} \leq L^+(u).$$

Thus, if we prove (point-to-prove), then $\Lambda = L^+(u)$, which is exactly the conclusion of Theorem 6.2.

We will obtain (point-to-prove) in Proposition 6.7 below. Towards that end, we will prove two special cases of (point-to-prove), which are covered by Lemma 6.5 and Lemma 6.6.

Lemma 6.5

Let $u_1 \leq u_2 \leq \cdots \leq u_n \leq \cdots$ be in $\mathrm{Bor}^+(X, \mathbb{R})$ and let $A \in \mathfrak{M}$ be such that $u_n \nearrow \chi_A$. Consider the increasing limit

$$\Lambda := \lim_{n\to\infty} L^+(u_n) \in [0, \infty].$$

Then $\Lambda \geq \mu(A)$, where we note that $\mu(A) = L^+(\chi_A)$.

Proof: Assume $A \neq \varnothing$, since if $A = \varnothing$, then $\mu(A) = 0$ and there is nothing to prove. Fix for the moment a $\theta \in (0, 1)$. For every $n \in \mathbb{N}$, define

$$A_n := \{x \in A \mid u_n(x) > \theta\}$$

<u>Observe:</u>

1. $A_n \in \mathfrak{M}$ for all $n \in \mathbb{N}$. This is because $A_n = A \cap u_n^{-1}((\theta, \infty))$ with $A, u_n^{-1}((\theta, \infty)) \in \mathfrak{M}$.
2. $A_n \subseteq A_{n+1}$ for all $n \in \mathbb{N}$. Indeed, if $x \in A_n$, then $u_n(x) > \theta$, and since $u_n \leq u_{n+1}$, we get $u_{n+1}(x) > \theta$, so $x \in A_{n+1}$.
3. $\bigcup_{n=1}^{\infty} A_n = A$. Indeed, $A_n \subseteq A$ for all $n \in \mathbb{N}$, hence $\bigcup_{n=1}^{\infty} A_n \subseteq A$. For "$\supseteq$", pick $x \in A$. Since $u_n \nearrow \chi_A$, we have $u_n(x) \to \chi_A(x) = 1$. Since $\theta < 1$, there exists $n \in \mathbb{N}$ such that $u_n(x) > \theta$, hence $x \in A_n$.

Thus $(A_n)_{n=1}^{\infty}$ is an increasing chain of sets in $\mathfrak{M}$ with $\bigcup_{n=1}^{\infty} A_n = A$. Using Proposition 2.12, we have

$$\lim_{n\to\infty} \mu(A_n) = \mu(A) \qquad (*)$$

Now observe that for every $n \in \mathbb{N}$ we have $u_n \geq \theta\chi_{A_n}$. Indeed, if $x \notin A_n$, then $\theta\chi_{A_n}(x) = 0 \leq u_n(x)$, while if $x \in A_n$, then $\theta\chi_{A_n}(x) = \theta < u_n(x)$.

Hence by Proposition 5.9, Proposition 5.8, and Theorem 5.2,

$$L^+(u_n) \geq L^+\left(\theta\chi_{A_n}\right) = \theta L^+\left(\chi_{A_n}\right) = \theta L_s^+\left(\chi_{A_n}\right) = \theta\mu(A_n).$$

So we have

$$\text{For every } n \in \mathbb{N},\ L^+(u_n) \geq \theta\mu(A_n) \tag{$**$}$$

Letting $n \to \infty$ in $(**)$, we get

$$\Lambda \geq \theta\mu(A)$$

where we use $(*)$ for the right-hand side. Finally, unfix θ. We have $\Lambda \geq \theta\mu(A)$ for every $\theta \in (0, 1)$, and hence $\Lambda \geq \mu(A)$, as desired. $\square$

Lemma 6.6

Let $u_1 \leq u_2 \leq \cdots \leq u_n \leq \cdots$ be in $\mathrm{Bor}^+(X, \mathbb{R})$ and let $f_0 \in \mathrm{Bor}_s^+(X, \mathbb{R})$ be such that $u_n \nearrow f_0$. Consider the increasing limit

$$\Lambda := \lim_{n\to\infty} L^+(u_n) \in [0, \infty].$$

Then $\Lambda \geq L_s^+(f_0)$, where we note that $L_s^+(f_0) = L^+(f_0)$.

Proposition 6.7

Let $u_n \nearrow u$ in $\mathrm{Bor}^+(X, \mathbb{R})$, and let us consider the increasing limit

$$\Lambda := \lim_{n\to\infty} L^+(u_n) \in [0, \infty].$$

Then $\Lambda \geq L^+(u)$.

Theorem 6.8 **Characterization of L^+**

The map $L^+ : \mathrm{Bor}^+(X, \mathbb{R}) \to [0, \infty]$ has the following properties, which determine it uniquely:

(i) $L^+(\chi_A) = \mu(A)$ for every $A \in \mathfrak{M}$.
(ii) $L^+(u + v) = L^+(u) + L^+(v)$ for every $u, v \in \mathrm{Bor}^+(X, \mathbb{R})$.
(iii) $L^+(\alpha u) = \alpha \cdot L^+(u)$ for every $u \in \mathrm{Bor}^+(X, \mathbb{R})$ and $\alpha \in [0, \infty)$.
(iv) $\lim_{n\to\infty} L^+(u_n) = L^+(u)$ whenever u and $(u_n)_{n=1}^{\infty}$ are in $\mathrm{Bor}^+(X, \mathbb{R})$ and $u_n \nearrow u$.

Note

For the existence part in Theorem 6.8, property (iv) is Theorem 6.2, while properties (i), (ii), (iii) follow from Theorem 5.2, combined with the fact that L^+ is an extension of the map $L_s^+ : \mathrm{Bor}_s^+(X, \mathbb{R}) \to [0, \infty]$.

For the uniqueness part in Theorem 6.8, it is convenient to record the following lemma, which will also turn out to be useful in subsequent lectures.

Lemma 6.9 **Method of the friendly functions**

Let $(X, \mathfrak{M})$ be a measurable space and consider a set of functions $\mathcal{G} \subseteq \mathrm{Bor}^+(X, \mathbb{R})$ which has the properties listed below.

(i) $\chi_A \in \mathcal{G}$ for every $A \in \mathfrak{M}$.
(ii) If $g_1, g_2 \in \mathcal{G}$, then $g_1 + g_2 \in \mathcal{G}$.
(iii) If $g \in \mathcal{G}$ and $\alpha \in [0, \infty)$, then $\alpha g \in \mathcal{G}$.
(iv) If $g \in \mathrm{Bor}^+(X, \mathbb{R})$, $g_1 \leq g_2 \leq \cdots \leq g_n \leq \cdots$ in $\mathcal{G}$, and $g_n \nearrow g$, then $g \in \mathcal{G}$.

Under these hypotheses, it follows that $\mathcal{G} = \mathrm{Bor}^+(X, \mathbb{R})$.

Proof:

- First, every $f \in \mathrm{Bor}_s^+(X, \mathbb{R})$ belongs to $\mathcal{G}$. Indeed, by Lemma 5.4, we can write

$$f = \alpha_1 \chi_{A_1} + \cdots + \alpha_n \chi_{A_n}$$

  with $A_1, \ldots, A_n \in \mathfrak{M}$ and $\alpha_1, \ldots, \alpha_n \in [0, \infty)$. By properties (i), (ii), and (iii), it follows that $f \in \mathcal{G}$.
- Now let $u \in \mathrm{Bor}^+(X, \mathbb{R})$. By Proposition 4.7, there exists a sequence $(f_n)_{n=1}^{\infty}$ in $\mathrm{Bor}_s^+(X, \mathbb{R})$ such that $f_n \nearrow u$. By the previous bullet, $f_n \in \mathcal{G}$ for every $n \in \mathbb{N}$, and hence property (iv) gives $u \in \mathcal{G}$.

$\square$

Proof of uniqueness in Theorem 6.8: Suppose we are given a map

$$M^+ : \mathrm{Bor}^+(X, \mathbb{R}) \to [0, \infty]$$

which also satisfies the conditions (i)-(iv) in Theorem 6.8. Let

$$\mathcal{G} = \{g \in \mathrm{Bor}^+(X, \mathbb{R}) \mid L^+(g) = M^+(g)\} \subseteq \mathrm{Bor}^+(X, \mathbb{R})$$

We will show that $\mathcal{G} = \mathrm{Bor}^+(X, \mathbb{R})$ by checking the hypotheses of Lemma 6.9.

- If $A \in \mathfrak{M}$, then $\chi_A \in \mathcal{G}$ because

$$L^+(\chi_A) = \mu(A) = M^+(\chi_A)$$

- If $g_1, g_2 \in \mathcal{G}$, then $g_1 + g_2 \in \mathcal{G}$ because

$$L^+(g_1 + g_2) = L^+(g_1) + L^+(g_2) = M^+(g_1) + M^+(g_2) = M^+(g_1 + g_2).$$

- If $g \in \mathcal{G}$ and $\alpha \in [0, \infty)$, then $\alpha g \in \mathcal{G}$ because

$$L^+(\alpha g) = \alpha L^+(g) = \alpha M^+(g) = M^+(\alpha g).$$

- Suppose $g \in \mathrm{Bor}^+(X, \mathbb{R})$, $g_1 \leq g_2 \leq \cdots \leq g_n \leq \cdots$ in $\mathcal{G}$, and $g_n \nearrow g$. Then

$$L^+(g) = \lim_{n\to\infty} L^+(g_n) = \lim_{n\to\infty} M^+(g_n) = M^+(g),$$

so $g \in \mathcal{G}$.

Thus $\mathcal{G} = \mathrm{Bor}^+(X,\mathbb{R})$ by Lemma 6.9. Hence $L^+(g) = M^+(g)$ for every $g \in \mathrm{Bor}^+(X,\mathbb{R})$, so L^+ is uniquely determined. □

7 The space of integrable functions $\mathcal{L}^1(\mu)$

7.1 Notion of integrable function, the map $L : \mathcal{L}^1(\mu) \to \mathbb{R}$

Remark 7.1 **Framework**

In this part, we continue to work with the fixed measure space $(X, \mathfrak{M}, \mu)$ from the previous chapter. There we considered

$$\mathrm{Bor}^+(X,\mathbb{R}) = \{f \in \mathrm{Bor}(X,\mathbb{R}) \mid f(x) \geq 0 \ \forall x \in X\}$$

and defined a map $L^+ : \mathrm{Bor}^+(X,\mathbb{R}) \to [0,\infty]$, which is trying to become the integral with respect to μ.

We now want to work with the full space $\mathrm{Bor}(X,\mathbb{R})$. The idea is to reduce a function $f \in \mathrm{Bor}(X,\mathbb{R})$ to a pair of functions $f_+, f_- \in \mathrm{Bor}^+(X,\mathbb{R})$, called the *positive part* and *negative part* of f, defined by

$$f_+ := f \vee \underline{0} \quad \text{and} \quad f_- := (-f) \vee \underline{0}.$$

Here $\vee$ denotes the pointwise maximum of two functions. Equivalently, for every $x \in X$,

$$f_+(x) = \begin{cases} f(x) & \text{if} \ \ f(x) \geq 0 \\ 0 & \text{if} \ \ f(x) < 0 \end{cases} \quad \text{and} \quad f_-(x) = \begin{cases} 0 & \text{if} \ \ f(x) \geq 0 \\ -f(x) & \text{if} \ \ f(x) < 0 \end{cases}.$$

From these formulas we have

$$f_+ + f_- = |f| \quad \text{and} \quad f_+ - f_- = f.$$

We also have the useful expressions

$$f_+ = \frac{f + |f|}{2} \quad \text{and} \quad f_- = \frac{|f| - f}{2}.$$

Lemma 7.2

For $f \in \mathrm{Bor}(X,\mathbb{R})$, we have the equivalence

$$L^+(|f|) < \infty \iff (L^+(f_+) < \infty \text{ and } L^+(f_-) < \infty).$$

Proof: This is immediate from the fact that

$$L^+(|f|) = L^+(f_+) + L^+(f_-),$$

which follows from $|f| = f_+ + f_-$ and the additivity of L^+ from Theorem 6.8. □

Definition 7.3 **Integrable functions and the map L**

A function $f : X \to \mathbb{R}$ is said to be *integrable with respect to* μ if $f \in \mathrm{Bor}(X, \mathbb{R})$ and the equivalent conditions from Lemma 7.2 hold. We denote

$$\mathcal{L}^1(\mu) := \{f : X \to \mathbb{R} \mid f \text{ is integrable with respect to } \mu\}.$$

We define a map $L : \mathcal{L}^1(\mu) \to \mathbb{R}$ by putting, for every $f \in \mathcal{L}^1(\mu)$,

$$L(f) := L^+(f_+) - L^+(f_-).$$

Note

We will prove in Theorem 7.5 that the function L is linear. Towards that end, we first prove a lemma which gives some leeway in how $L(f)$ is calculated: instead of having to stick with $f = f_+ - f_-$, the lemma allows us to use a writing $f = h_1 - h_2$.

Lemma 7.4

Let $f \in \mathcal{L}^1(\mu)$ and suppose that we can write $f = h_1 - h_2$ with $h_1, h_2 \in \mathrm{Bor}^+(X, \mathbb{R})$ such that $L^+(h_1), L^+(h_2) < \infty$. Then

$$L(f) = L^+(h_1) - L^+(h_2).$$

Proof: Since $f = h_1 - h_2$ and $f = f_+ - f_-$, we have

$$h_1 + f_- = h_2 + f_+.$$

Applying L^+ to both sides, and using additivity from Theorem 6.8, gives

$$L^+(h_1) + L^+(f_-) = L^+(h_2) + L^+(f_+).$$

All four quantities appearing here are finite, so we can rearrange this equality to obtain

$$L^+(f_+) - L^+(f_-) = L^+(h_1) - L^+(h_2).$$

The left-hand side is $L(f)$ by definition, and the result follows. □

Theorem 7.5 **Linearity of L**

1. $\mathcal{L}^1(\mu)$ is stable under linear combinations, hence is a linear subspace of $\mathrm{Bor}(X, \mathbb{R})$.
2. The map $L : \mathcal{L}^1(\mu) \to \mathbb{R}$ is linear.

Proof of 1: Pick some $f, g \in \mathcal{L}^1(\mu)$ and $\alpha, \beta \in \mathbb{R}$, and form the linear combination

$$h := \alpha f + \beta g \in \mathrm{Bor}(X, \mathbb{R}).$$

We want to prove that $h \in \mathcal{L}^1(\mu)$. Observe that

$$|h| = |\alpha f + \beta g| \leq |\alpha| \cdot |f| + |\beta| \cdot |g|,$$

which implies that

$$\begin{aligned} L^+(|h|) &\leq L^+(|\alpha| \cdot |f| + |\beta| \cdot |g|) \\ &= |\alpha| \cdot L^+(|f|) + |\beta| \cdot L^+(|g|) \\ &< \infty. \end{aligned}$$

The first inequality follows from the fact that L^+ is increasing, and the second equality follows from the additivity and homogeneity properties in Theorem 6.8. Since $f, g \in \mathcal{L}^1(\mu)$, the last expression is finite. It follows that $L^+(|h|) < \infty$, so $h \in \mathcal{L}^1(\mu)$ by Lemma 7.2. □

Proof of 2: We will verify separately the two statements which, together, give linearity:

(a) $L(f+g) = L(f) + L(g)$ for every $f, g \in \mathcal{L}^1(\mu)$.
(b) $L(\alpha f) = \alpha L(f)$ for every $f \in \mathcal{L}^1(\mu)$ and $\alpha \in \mathbb{R}$.

Verification of (a):
Pick $f, g \in \mathcal{L}^1(\mu)$. By 1, $f + g \in \mathcal{L}^1(\mu)$. We can write

$$f + g = (f_+ + g_+) - (f_- + g_-),$$

where $f_+ + g_+$ and $f_- + g_-$ are in $\mathrm{Bor}^+(X, \mathbb{R})$. Moreover,

$$L^+(f_+ + g_+) = L^+(f_+) + L^+(g_+) < \infty$$

and

$$L^+(f_- + g_-) = L^+(f_-) + L^+(g_-) < \infty,$$

since $f, g \in \mathcal{L}^1(\mu)$. Thus Lemma 7.4 gives

$$\begin{aligned} L(f+g) &= L^+(f_+ + g_+) - L^+(f_- + g_-) \\ &= L^+(f_+) + L^+(g_+) - L^+(f_-) - L^+(g_-) \\ &= L(f) + L(g). \end{aligned}$$

Verification of (b):
Pick $f \in \mathcal{L}^1(\mu)$ and $\alpha \in \mathbb{R}$. By 1, $\alpha f \in \mathcal{L}^1(\mu)$.

If $\alpha \geq 0$, then

$$\alpha f = \alpha f_+ - \alpha f_-,$$

and $L^+(\alpha f_+) = \alpha L^+(f_+) < \infty$, $L^+(\alpha f_-) = \alpha L^+(f_-) < \infty$. Hence Lemma 7.4 gives

$$L(\alpha f) = \alpha L^+(f_+) - \alpha L^+(f_-) = \alpha L(f).$$

If $\alpha < 0$, then

$$\alpha f = (-\alpha) f_- - (-\alpha) f_+,$$

and $L^+((-\alpha) f_-) = (-\alpha) L^+(f_-) < \infty$, $L^+((-\alpha) f_+) = (-\alpha) L^+(f_+) < \infty$. Hence Lemma 7.4 gives

$$L(\alpha f) = (-\alpha) L^+(f_-) - (-\alpha) L^+(f_+) = \alpha L(f).$$

□

Proposition 7.6

1. If $f \in \mathcal{L}^1(\mu)$ and also $f \in \mathrm{Bor}^+(X, \mathbb{R})$, then $L(f) = L^+(f)$.
2. If $f, g \in \mathcal{L}^1(\mu)$ are such that $f \leq g$, then $L(f) \leq L(g)$.
3. If $f \in \mathcal{L}^1(\mu)$, then $|f| \in \mathcal{L}^1(\mu)$ as well, and

$$|L(f)| \leq L(|f|).$$

Proof of 1: Since $f \in \mathrm{Bor}^+(X, \mathbb{R})$, we have $f_+ = f$ and $f_- = \underline{0}$. Hence

$$L(f) = L^+(f_+) - L^+(f_-) = L^+(f) - L^+(\underline{0}) = L^+(f).$$

$\square$

Proof of 2: If $f \leq g$, then $g - f \in \mathcal{L}^1(\mu)$ by Theorem 7.5, and moreover $g - f \in \mathrm{Bor}^+(X, \mathbb{R})$. By part 1,

$$L(g - f) = L^+(g - f) \geq 0.$$

Using the linearity of L, this says that $L(g) - L(f) \geq 0$, hence $L(f) \leq L(g)$. $\square$

Proof of 3: Since $f \in \mathcal{L}^1(\mu)$, Lemma 7.2 gives $L^+(|f|) < \infty$, so $|f| \in \mathcal{L}^1(\mu)$.

As functions, we have

$$-|f| \leq f \leq |f|.$$

Applying part 2 gives

$$-L(|f|) \leq L(f) \leq L(|f|),$$

where we also used the linearity of L on the left. Thus

$$|L(f)| \leq L(|f|).$$

$\square$

Remark 7.7 **Switching to integral notation**

For our measure space $(X, \mathfrak{M}, \mu)$, we have now constructed two maps

$$L^+ : \mathrm{Bor}^+(X, \mathbb{R}) \to [0, \infty] \quad \text{and} \quad L : \mathcal{L}^1(\mu) \to \mathbb{R},$$

which are trying to become "the integral with respect to μ". If $f \in \mathrm{Bor}(X, \mathbb{R})$ is such that both values $L^+(f)$ and $L(f)$ are defined, then Proposition 7.6 says that $L(f) = L^+(f)$.

It is therefore safe to switch to the customary integral notation:

$$\text{if at least one of } \; L^+(f), L(f) \; \text{ is defined, we denote that value by}$$
$$\int_X f(x)\,\mathrm{d}\mu(x).$$

This is often abbreviated to just $\int f\,\mathrm{d}\mu$.

While we now have uniform notation for all our integrals, it is useful to remember that the integral notation can have two meanings:
- an L^+ meaning, when it has values in $[0, \infty]$ and we use the properties of L^+;
- an L meaning, when it has values in $\mathbb{R}$ and we use the properties of L.

This usually causes no harm. For instance, the inequality from Proposition 7.6 becomes

$$\left| \int_X f(x)\,\mathrm{d}\mu(x) \right| \le \int_X |f(x)|\,\mathrm{d}\mu(x), \quad \forall f \in \mathcal{L}^1(\mu),$$

where the integral on the right-hand side can be viewed either in the L^+ sense or in the L sense.

7.2 Integration of subsets

Notation 7.8

Let $(X, \mathfrak{M}, \mu)$ be a measure space.

1. For any $f \in \mathrm{Bor}^+(X, \mathbb{R})$ and $A \in \mathfrak{M}$, we denote

$$\int_A f\,\mathrm{d}\mu := \int_X f \cdot \chi_A\,\mathrm{d}\mu \in [0, \infty],$$

where, as usual, χ_A denotes the indicator function of A.

2. For any $f \in \mathcal{L}^1(\mu)$ and $A \in \mathfrak{M}$, observe that $f \cdot \chi_A \in \mathcal{L}^1(\mu)$, since $|f \cdot \chi_A| \le |f|$. We denote

$$\int_A f\,\mathrm{d}\mu := \int_X f \cdot \chi_A\,\mathrm{d}\mu \in \mathbb{R}.$$

Proposition 7.9

Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $h \in \mathrm{Bor}^+(X, \mathbb{R})$. We define a set-function $\nu : \mathfrak{M} \to [0, \infty]$ by putting

$$\nu(A) := \int_A h \, \mathrm{d}\mu, \quad A \in \mathfrak{M}.$$

Then ν is a positive measure on $\mathfrak{M}$.

Proof:

1. $$\nu(\varnothing) = \int_X h \cdot \chi_{\varnothing} \, \mathrm{d}\mu = \int_X \underline{0} \, \mathrm{d}\mu = 0.$$
2. Fix $(A_n)_{n=1}^{\infty}$ in $\mathfrak{M}$ such that $A_i \cap A_j = \varnothing$ for $i \neq j$, and let $U = \bigcup_{n=1}^{\infty} A_n$. We must prove that

$$\nu(U) = \sum_{n=1}^{\infty} \nu(A_n).$$

For every $N \in \mathbb{N}$, let

$$s_N = \sum_{n=1}^{N} h \cdot \chi_{A_n}.$$

Then $(s_N)_{N=1}^{\infty}$ is an increasing sequence in $\mathrm{Bor}^+(X, \mathbb{R})$, and

$$s_N \to h \cdot \chi_U$$

pointwise. Hence, by Theorem 6.2,

$$\begin{aligned} \nu(U) &= \int_X h \cdot \chi_U \, \mathrm{d}\mu \\ &= \lim_{N \to \infty} \int_X s_N \, \mathrm{d}\mu \\ &= \lim_{N \to \infty} \sum_{n=1}^{N} \int_X h \chi_{A_n} \, \mathrm{d}\mu \\ &= \sum_{n=1}^{\infty} \nu(A_n). \end{aligned}$$

□

Proposition 7.10

Let $(X, \mathfrak{M}, \mu)$, $h \in \mathrm{Bor}^+(X, \mathbb{R})$, and the positive measure $\nu : \mathfrak{M} \to [0, \infty]$ be as in Proposition 7.9. Then, for every $f \in \mathrm{Bor}^+(X, \mathbb{R})$,

$$\int_X f \, \mathrm{d}\nu = \int_X f \cdot h \, \mathrm{d}\mu \in [0, \infty].$$

This is an equality of integrals considered in the L^+ sense.

Proof: We will use the method of the friendly functions (Lemma 6.9). Let

$$\mathcal{G} = \left\{ g \in \mathrm{Bor}^+(X, \mathbb{R}) \;\middle|\; \int_X g \, \mathrm{d}\nu = \int_X g \cdot h \, \mathrm{d}\mu \right\}$$

We verify that $\mathcal{G}$ satisfies the conditions of Lemma 6.9.

- For $A \in \mathfrak{M}$, we have
$$\int_X \chi_A \, \mathrm{d}\nu = \nu(A) = \int_A h \, \mathrm{d}\mu = \int_X \chi_A \cdot h \, \mathrm{d}\mu,$$
so $\chi_A \in \mathcal{G}$.
- If $g_1, g_2 \in \mathcal{G}$, then
$$\begin{aligned} \int_X (g_1 + g_2) \, \mathrm{d}\nu &= \int_X g_1 \, \mathrm{d}\nu + \int_X g_2 \, \mathrm{d}\nu \\ &= \int_X g_1 \cdot h \, \mathrm{d}\mu + \int_X g_2 \cdot h \, \mathrm{d}\mu \\ &= \int_X (g_1 + g_2) \cdot h \, \mathrm{d}\mu, \end{aligned}$$
so $g_1 + g_2 \in \mathcal{G}$.
- If $g \in \mathcal{G}$ and $\alpha \in [0, \infty)$, then
$$\begin{aligned} \int_X \alpha g \, \mathrm{d}\nu &= \alpha \int_X g \, \mathrm{d}\nu \\ &= \alpha \int_X g \cdot h \, \mathrm{d}\mu \\ &= \int_X (\alpha g) \cdot h \, \mathrm{d}\mu, \end{aligned}$$
so $\alpha g \in \mathcal{G}$.
- Suppose that $g_1 \le g_2 \le \cdots \le g_n \le \cdots$ are in $\mathcal{G}$ and $g_n \to g$ for some $g \in \mathrm{Bor}^+(X, \mathbb{R})$. Since $h \ge 0$, we also have $g_n \cdot h \to g \cdot h$ increasingly. By Theorem 6.2, applied once with respect to ν and once with respect to μ,

$$\begin{aligned}\int_X g \, \mathrm{d}\nu &= \lim_{n\to\infty} \int_X g_n \, \mathrm{d}\nu \\ &= \lim_{n\to\infty} \int_X g_n \cdot h \, \mathrm{d}\mu \\ &= \int_X g \cdot h \, \mathrm{d}\mu.\end{aligned}$$

Thus $g \in \mathcal{G}$.

Then Lemma 6.9 gives $\mathcal{G} = \mathrm{Bor}^+(X, \mathbb{R})$, which means that

$$\int_X g \, \mathrm{d}\nu = \int_X g \cdot h \, \mathrm{d}\mu$$

for all $g \in \mathrm{Bor}^+(X, \mathbb{R})$, as required. □

Note

In connection with Proposition 7.10, it is customary to write the connection between the positive measures μ and ν in the form

$$\mathrm{d}\nu = h \, \mathrm{d}\mu.$$

This can be thought of as cancelling the common integral expression in

$$\int_X f \, \mathrm{d}\nu = \int_X f \cdot h \, \mathrm{d}\mu.$$

Some further abuse of notation rewrites this relation as

$$h = \frac{\mathrm{d}\nu}{\mathrm{d}\mu},$$

where one thinks of h as a kind of “derivative” of ν with respect to μ.

8 Lebesgue’s Dominated Convergence Theorem (LDCT)

Fix a measure space $(X, \mathfrak{M}, \mu)$.

Theorem 8.1 **LDCT**

Let f and $(f_n)_{n=1}^{\infty}$ be functions in $\mathrm{Bor}(X, \mathbb{R})$ such that

$$\lim_{n\to\infty} f_n(x) = f(x), \quad \forall x \in X.$$

Suppose moreover that there exists a function $h \in \mathcal{L}^1(\mu) \cap \mathrm{Bor}^+(X, \mathbb{R})$ which dominates all the f_n, in the sense that

$$|f_n| \leq h, \quad \forall n \in \mathbb{N}.$$

Then f and $f_1, f_2, \ldots, f_n, \ldots$ are all in $\mathcal{L}^1(\mu)$, and

$$\lim_{n\to\infty} \int_X f_n \, \mathrm{d}\mu = \int_X f \, \mathrm{d}\mu.$$

Note

The last equation can be written as

$$\lim_{n\to\infty} \left(\int_X f_n \, \mathrm{d}\mu \right) = \int_X \left(\lim_{n\to\infty} f_n \right) \mathrm{d}\mu,$$

so this is a formula where one interchanges integration with a limit. In principle, this interchange can have issues. The point of LDCT is that the existence of the dominating function h is a useful sufficient condition which makes the issues go away.

It is convenient to think of LDCT as going in parallel with the MCT from Theorem 6.2. In connection to that, we will observe in Proposition 8.2 that a special case of LDCT gives a "Reverse-MCT" statement.

Proposition 8.2 **Reverse-MCT**

Let u and $(u_n)_{n=1}^{\infty}$ be functions in $\mathrm{Bor}^+(X, \mathbb{R})$ such that

$$u_1 \geq u_2 \geq \cdots \geq u_n \geq \cdots$$

and such that

$$\lim_{n\to\infty} u_n(x) = u(x), \quad \forall x \in X.$$

(compactly written $u_n \searrow u$). We assume, in addition, that $\int_X u_1 \, \mathrm{d}\mu < \infty$. Then

$$\lim_{n\to\infty} \int_X u_n \, \mathrm{d}\mu = \int_X u \, \mathrm{d}\mu.$$

Proof: Define

$$v_n := u_1 - u_n, \quad n \in \mathbb{N}, \quad \text{and} \quad v := u_1 - u.$$

Then $v_n \in \mathrm{Bor}^+(X, \mathbb{R})$ for every n, and $v_n \nearrow v$. By Theorem 6.2,

$$\lim_{n\to\infty} \int_X v_n \, \mathrm{d}\mu = \int_X v \, \mathrm{d}\mu.$$

Since $u_1 = v_n + u_n$ and $u_1 = v + u$, and since $\int_X u_1 \, \mathrm{d}\mu < \infty$, all the quantities below are finite and we can write

$$\begin{aligned}\int_X u_n \, \mathrm{d}\mu &= \int_X u_1 \, \mathrm{d}\mu - \int_X v_n \, \mathrm{d}\mu \\ &\to \int_X u_1 \, \mathrm{d}\mu - \int_X v \, \mathrm{d}\mu \\ &= \int_X u \, \mathrm{d}\mu.\end{aligned}$$

□

Note

Reverse-MCT is a special case of LDCT because we can use $h = u_1$ as the dominating function. Then Theorem 8.1 gives the required conclusion.

For the plan of how we prove LDCT, it is convenient to also consider its special case LDCT_0, which considers a dominated sequence in $\mathrm{Bor}^+(X, \mathbb{R})$ converging pointwise to 0. This is Proposition 8.3.

Proposition 8.3 LDCT_0

Let $(g_n)_{n=1}^\infty$ be a sequence of functions in $\mathrm{Bor}^+(X, \mathbb{R})$ such that

$$\lim_{n\to\infty} g_n(x) = 0, \quad \forall x \in X.$$

Suppose moreover that there exists a function $h \in \mathcal{L}^1(\mu) \cap \mathrm{Bor}^+(X, \mathbb{R})$ such that

$$g_n \le h, \quad \forall n \in \mathbb{N}.$$

Then

$$\lim_{n\to\infty} \int_X g_n \, \mathrm{d}\mu = 0.$$

Proof: The idea is to make a helper sequence $(u_n)_{n=1}^\infty$ to which we can apply Proposition 8.2. We break the proof into steps:

1. For every $n \in \mathbb{N}$, define $u_n : X \to [0, \infty)$ by

   $$u_n(x) := \sup\{g_k(x) \mid k \ge n\}.$$

   Since each $g_k \in \mathrm{Bor}^+(X, \mathbb{R})$, Proposition 4.1 gives $u_n \in \mathrm{Bor}(X, \mathbb{R})$. Also $0 \le u_n \le h$, so $u_n \in \mathrm{Bor}^+(X, \mathbb{R})$. In particular,

$$\int_X u_1 \, \mathrm{d}\mu \leq \int_X h \, \mathrm{d}\mu < \infty,$$

where the first inequality uses the monotonicity of L^+ from Theorem 6.8, and the finiteness of the last term follows from Proposition 7.6. Finally, from the definition of the tail suprema we have

$$u_1 \geq u_2 \geq \cdots \geq u_n \geq \cdots.$$

2. We claim that $u_n \searrow \underline{0}$. Fix $x \in X$ and let $\varepsilon > 0$ be given. Since $g_n(x) \to 0$, there is $n_0 \in \mathbb{N}$ such that $g_k(x) < \varepsilon$ for all $k \geq n_0$. For this n_0, we have

$$u_{n_0}(x) = \sup\{g_k(x) \mid k \geq n_0\} \leq \varepsilon.$$

Since the sequence $(u_n(x))_{n=1}^{\infty}$ is decreasing, it follows that

$$0 \leq u_n(x) \leq u_{n_0}(x) \leq \varepsilon, \quad \forall n \geq n_0.$$

Thus $u_n(x) \to 0$ for every $x \in X$.

3. Now Proposition 8.2 applied to $(u_n)_{n=1}^{\infty}$ gives

$$\lim_{n \to \infty} \int_X u_n \, \mathrm{d}\mu = 0.$$

Since $g_n \leq u_n$ for every $n \in \mathbb{N}$, the monotonicity of L^+ gives

$$0 \leq \int_X g_n \, \mathrm{d}\mu \leq \int_X u_n \, \mathrm{d}\mu, \quad \forall n \in \mathbb{N}.$$

Hence $\int_X g_n \, \mathrm{d}\mu \to 0$ as required.

$\square$